Statistical Analysis of Dark Matter: XENON1T S2-only experiment

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Abstract

This paper presents a re-analysis of the XENON1T S2-only search for light dark matter using both Bayesian and Frequentist statistical frameworks to analyze the background and set upper limits on light dark matter interactions. For the analysis the public data from the XENON1T S2-only science run (SR1) was used, which covered 258.2 live days and 0.977 tonne year raw exposure. Key imported files include observed ionisation events, detector response matrices, bin definitions, and nominal ER/CEvNS background models. A binned Poisson likelihood with two free parameters(background normalisation β (nuisance) and signal strength μ_s) is implemented. The background is fitted with both frequentist (MLE + curvature uncertainty) and Bayesian methods (posterior under flat and scale-invariant priors). Signal upper limits are derived using the standard Poisson method (implementing Brent root-finding), and model comparison uses Bayes factors. The background normalisation converges to β equal to 7.6 across all methods, indicating the nominal model under-predicts the data by a factor of 7-8. No significant DM signal was found; a 90% CL upper limit on the DM nucleon cross section is computed using the standard Poisson method, yielding $\sigma < 3.72 \times 10^{-43}$ cm^2 at $m_\chi = 4$ GeV/c^2. Bayes factors $B_{01} < 1$ favour the signal hypothesis, although this is likely driven by the incomplete background model rather than genuine DM evidence.

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1 Introduction

Astronomical phenomena, such as galaxy rotation curves and gravitational lensing, provide strong evidence that the Universe is made of non-luminous matter that does not interact electromagnetically with ordinary barionic matter. However, its' fundamental nature remains unknown [1]. The study of dark matter (DM) underlies both the cosmology and particle physics fields, as it makes up a significant fraction of the total mass and energy of the Universe and could be explained by an undiscovered subatomic particle. Weakly Interacting Massive Particles (WIMPS) are one of the candidates.[2]

This project studies the paper "Light Dark Matter Search with Ionization Signals in XENON1T", which aims to observe dark matter particles via their interaction with xenon atoms in a detector. This detector is located in the INFN Laboratori Nazionali del Gran Sasso, and consists of a 2 ton liquid xenon chamber. It is surrounded by an electrode at the top and a cathode at the bottom, with a separation of 97cm. Inside, electrons and photons are produced by charged particles, which are guided by electric fields and then measured by photomultiplier tubes for the acquisition of data .

The observable quantity in this experiment is therefore the rate and energy of ionization events, expressed as the number of deposited electrons or their corresponding energy (PE). The dominant background contributions can be attributed to electronic (er) and nuclear recoil (nr). By comparing the observed events and filtering the background fit, we could obtain new information about WIMPS and their properties.

This report aims to follow up on the data analysis of the XENON1T paper. In order to do so, first, the physical background of the experiment is introduced and the data imports are explained. Then, some relevant theory about the model, the likelihood and priors are introduced. Finally, the inference procedure is explained before analyzing the results and the corresponding discussion.

2 Physical Background

The detection of DM in the XENON1T experiment relies on a dual-phase TPC. The detection occurs when a particle interacts with the liquid xenon (LXe). This deposit of energy mass produces two distinct signals. The first is a scintillation signal (S1) that the LXe radiates after absorbing the particle. The second is a delayed ionisation signal (S2) that releases electrons.

2.1 S2 Only analysis

The standard WIMP apparatus requires both signals to determine the coordinates of the particle interaction. However, light DM particles ($m_{DM} < 6$ GeV/c^2) lack the kinetic energy to scintillate and produce a detectable S1 signal.

Therefore, the coordinates of the DM events in the chamber can only be determined with an S2 signal that has a significantly lower detector threshold. This allows the experiment to detect nuclear recoils around 0.7 keV and electronic recoils around 0.186 keV. This smaller window allows the search for minute particle candidates such as light DM [3].

2.2 Background Components

It is worth mentioning that the data acquired during the experiment have been subject to many detection cuts due to physical limitations of the apparatus. An example of these experimental shortcomings is the non-negligible radiation of the chamber walls. Particle events occurring near the container have their ionization waves modified. These interferences render the data around the boundary regions unusable. Another example of necessary cuts is the Hit Pattern Cut. This cut aims to ignore the spontaneous release of electrons in Xenon gas liquid interface. If multiple random emissions occur in a small time frame, their individual signals interfere constructively and resemble a S2 signal.

While the aforementioned cuts define a clean volume of the detector, they cannot eliminate background events that originate from legitimate particle interactions within that volume. These physical background produce S2 signals that are indistinguishable from WIMP nuclear scattering, acting as a false positive

in the search of DM [3]. The primary components are the following:

1. Flat-spectrum Electronic Recoils (ER) arises from β-decays of ^{214}Pb. Because these decays have high energy mass compared to DM, they create a flat spectrum across the S2 bins.
2. Coherent Elastic Neutrino Nucleus Scattering (CEvNS) is caused by solar neutrinos scattering on XLe nuclei which creates a signal that falls in the 6 GeV/c^2 mass range; this mass can be wrongly interpreted as light DM. These low interaction particles have high penetrative properties meaning this background cannot be shielded from the photoelectric detectors.
3. Cathode Events are radioactive decays occurring on the cathode wires themselves at the bottom of the chamber. The electric field causes charge loss which causes high energy events to appear smaller and therefore cause false positives in DM detection.

To account for the cumulative impact of these physical processes, we introduce a single nuisance parameter, β, which servers to normalise the total expected background model (b_i) against the observed data as discussed in the Model section.

3 Data

The data used in this analysis was taken from the public data release accompanying the XENON1T S2-only dark matter research([3]). The data set corresponds to the main science run(SR1) of XENON1T with a live time of 258.2 days. The total raw exposure used in the analysis was 0.97678 tonne year; the effective one is much lower though, due to strong event selection.
The primary files are:

- **events_after_cuts.csv**: in this file each row is one event (candidate interaction) and the columns are quantities measured for that event. The most relevant for the analysis is:
  - **s2_area_pe**: a particle interaction deposits energy and ionisation electrons drift upward to be extracted into gas; each electron produces around 20 secondary photons. PMTs collect these photons. The integrated charge is what is recorded here.
- **s2_response_nr.csv**: the detector response in NR events is captured here. For the analysis we start from the 4th column (s2_bin_000) since the 3 previous ones are energies (not needed to get the expected number of counts).
- **s2_binning_info.csv**: this one gives logarithmically spaced bin edges from 90 to 3000 PE.
- **er_and_cevns_backgrounds.csv**: contains the number of expected background events in each S2 bin for the nominal ER and CEvNS background models. It is important to note that cathode background is not provided, which means the model is incomplete.

Regarding event selection, the authors applied a 0.7 keV energy cutoff for NR events. Besides, they chose an S2 region of interest which we also use, namely, the range (165.3,271.7) PE.
There are two relevant sources of uncertainty:

1. Statistical uncertainty intrinsic to the Poisson process itself.
2. Systematic uncertainty, which we model through the nuisance parameter β.

Each bin count k_i is realizationon of a Poisson random variable, because there is no statistical correlation between bins. This means no covariance matrix is needed.

4 Model

The model used for the analysis has two free parameters: the signal strength μ_s and the background normalization β. The signal strength describes how much dark matter signal is present in the data while the background normalization is a nuisance parameter that must be included so as to model uncertainty in the nominal background predictions(only ER and CEvNS are used, excluding cathode background).

The expected number of signals (in a given bin) coming from Dark Matter (DM) scattering events is given by:

$$s_i = \mathcal{E} \times \sum_i \left.\frac{dR}{dE}\right|_{E_i} \Delta E_i \, R_{ij} \tag{1}$$

Where $\mathcal{E}$ is the raw exposure (detector mass $\times$ running time), $\frac{dR}{dE}$ the theoretical event rate, ΔE_i the bin width and R_{ij} the response matrix.
This formula expresses the fact that the detector does not perfectly capture the true-energy spectrum, and instead, it may be the case that events who were truly in the range of bin i end up reconstructed in bin j. This detector response is captured in the response matrices (s2_response_er.csv, s2_response_nr.csv,). Different Dark Matter models will produce different event rates $\frac{dR}{dE}$, and they might offer a prediction for only nuclear recoils(nr) or only electronic recoils(er), which is why the authors of the paper divide the response matrices into two different files (one for "nr" and another one for "er").
For our analysis, same as in the example analysis by the authors, the Dark Matter model used is a 4 GeV/c^2 spin-independent elastic WIMP spectrum.
The expected number of counts per bin in this model is therefore:

$$\mu_i = \mu_s s_i + \beta b_i \tag{2}$$

where b_i is the nominal background.

5 Statistical Methods

In our analysis we use a combination of both Bayesian and Frequentist methods. Most of the differences between frequentist and Bayesian analyses are philosophical rather than quantitative. Frequentists consider parameters as fixed, unknown constants with only one true value, while Bayesian treat them as random variables. Frequentists ignore prior knowledge, preferring to work with confidence intervals, and infer uncertainties by how sharply the likelihood peaks. Bayesians, on the other hand, force the explicit inclusion of prior information and, to quantify uncertainty, prefer the credible interval, a range which holds a 68% chance of having the parameter in it. When priors are not informative and there is sufficient data, both analyses tend to agree. However, if a strongly informative prior(e.g. a tight prior centered around 1) the difference can become quite significant. [4]

5.1 Likelihood

The observed counts k_i in each S2 bin are independent Poisson random variables. The binned Poisson likelihood is:

$$\mathcal{L}(\mu_s, \beta) = \prod_{i=1}^{N} \frac{{\mu_i}^{k_i} e^{-\mu_i}}{k_i!}, \tag{3}$$

For convenience though, we will mostly use the log-likelihood:

$$\ln \mathcal{L}(\mu_s, \beta) = \sum_{i=1}^{N} \left[k_i \ln \mu_i - \mu_i - \ln(k_i!)\right] \tag{4}$$

As mentioned before, there is no covariance matrix since the bins are statistically independent.

5.2 Priors

We investigate different choices of prior to assess the sensitivity of the posterior to prior assumptions. For the background normalization β, the safest choice is to first try a scale-invariant or log-uniform prior of the form

$$p(\beta) \propto \frac{1}{\beta}, \beta > 0 \tag{5}$$

which reflects prior ignorance about the overall scale of the background. In practice, the prior covers a finite range such that $\beta \in [\beta_{min}, \beta_{max}]$.
As an alternative, one may consider a uniform prior,

$$p(\beta) = \frac{1}{\beta_{max} - \beta_{min}} \tag{6}$$

which assigns equal probability to all values within a fixed interval.
For the signal strength μ_s, we adopt a uniform prior over positive values

$$p(\mu_s) = const, \mu_s \geq 0 \tag{7}$$

reflecting the fact that the signal cannot be negative. These choices aim to remain minimally informative while incorporating basic physical constraints.

5.3 Bayes factor

We also compute a Bayes factor comparing the model 0 (background-only hypothesis) and the model 1 (which takes into account not only the background but also the signal of Dark Matter).

$$B_{01} = \frac{p(\text{data} \mid H_0)}{p(\text{data} \mid H_1)} \tag{8}$$

To calculate the evidence of each model ($p(\text{data} \mid H_0)$ and $p(\text{data} \mid H_1)$), we use the following equations:

$$p(\text{data} \mid H_0) = \int \mathcal{L}(\beta)\pi(\beta)\, d\beta \tag{9}$$

$$p(\text{data} \mid H_1) = \int\int \mathcal{L}(\beta, \mu_s)\pi(\beta, \mu_s)\, d\beta\, d\mu_s \tag{10}$$

Consequently, for the Bayes factor, we need to define a Bayesian likelihood that treats both the signal strength μ_s and background normalization β as random variables.

6 Inference Procedure

The aim of this section is to describe how the posterior distribution of parameters was obtained.

6.1 Background normalization β

To obtain the expected number of background events in bin i we simply set $\mu_s = 0$ in (2) and thus get:

$$\mu_i = \beta b_i \tag{11}$$

By adopting the Bayesian perspective, we take the nuisance parameter β to be a random variable. The goal at this point is to infer the posterior probability $p(\beta|data)$. For this we will need the likelihood (with $\mu_s = 0$) and a prior probability for β. Although the evidence ($P(data)$) will become important later when comparing models, we shall ignore it for now, since it is only a normalization constant independent of β. The posterior probability for the background parameter is therefore:

$$p(\beta|Data) \propto \mathcal{L}(\beta)p(\beta) \tag{12}$$

To simplify calculations we may take the natural logarithm of both sides:

$$\ln(p(\beta|Data)) = \ln(\mathcal{L}(\beta)) + \ln(p(\beta)) + const \tag{13}$$

The log likelihood for background only assuming a scale-invariant prior for β is:

$$\ln(\mathcal{L}(\beta)) = \ln(\beta)\sum_{i=1}^{N} k_i + \sum_{i=1}^{N} k_i \ln(b_i) - \beta\sum_{i=1}^{N} b_i - \sum_{i=1}^{N} \ln(k_i!) \tag{14}$$

Defining $K_{tot} = \sum_{i=1}^{N} k_i$ and $B_{tot} = \sum_{i=1}^{N} b_i$, we get:

$$\ln(\mathcal{L}(\beta)) = \ln(\beta)K_{tot} - \beta B_{tot} + const \tag{15}$$

Noting that $\ln(p(\beta)) = -\ln(\beta) + const$ the log posterior probability is:

$$\ln(p(\beta|Data)) = \ln(\beta)K_{tot} - \beta B_{tot} - \ln(\beta) + const \tag{16}$$

Simplifying:

$$\ln(p(\beta|Data)) = (K_{tot} - 1)\ln(\beta) - \beta B_{tot} + const \tag{17}$$

In the case of a flat prior:

$$\ln(p(\beta|Data) = ln(\beta)K_{tot} - \beta B_{tot} + \ln(p(\beta)) \tag{18}$$

Now we may sample from this posterior distribution using numerical methods and we can quantify uncertainties with the credible interval.

6.2 Signal strength μ_s

The signal strength (μ_s) is a dimensionless parameter, as previously defined in Equation 2. In order to analyze the background, we set this value to 0. For $\mu_s = 1$, however, we have a DM signal at the cross section $\sigma_{\text{ref}} = 10^{-45}$ cm^2. We can calculate an upper limit (μ_s^{up}) such that all $\mu_s > \mu_s^{\text{up}}$ are discarded at a specified confidence level. Once μ_s^{up} has been calculated, the upper limit on the physical cross section can be computed after proper scaling as follows:

$$\sigma^{\text{up}} = \mu_s^{\text{up}} \cdot \sigma_{\text{ref}} \tag{19}$$

The XENON1T paper uses the standard Poisson method that can be found in the PDG [5]. The paper computes a 90% confidence level upper limit, where μ_s^{up} is the value at which the probability of observing n_{obs} (or fewer) events is exactly 10%. This probability is given by the Poisson PDF [3]:

$$\text{CDF}(n_{\text{obs}} \mid \mu) = \sum_{k=0}^{n_{\text{obs}}} \frac{e^{-\mu}\mu^k}{k!} \tag{20}$$

where $\mu = \mu_s \cdot s^{\text{cons}} + \mu_b^{\text{cons}}$ is the total expected number of events at μ_s.
Equation 20 can't be solved analytically for μ_s, as this value appears inside the exponential and the power in all the terms in the sum. Therefore, we can solve it numerically using Brent's algorithm to find roots [6]. First, we define the following function:

$$f(\mu_s) = \sum_{k=0}^{n_{\text{obs}}} \frac{e^{-\mu}\,\mu^k}{k!} - 0.10 \tag{21}$$

and then we can find the value of μ_s where $f(\mu_s) = 0$. Since the function starts positive and ends negative, it crosses zero once. This point can be labeled μ_s^{up}.

7 Results

7.1 The background fit: the frequentist perspective

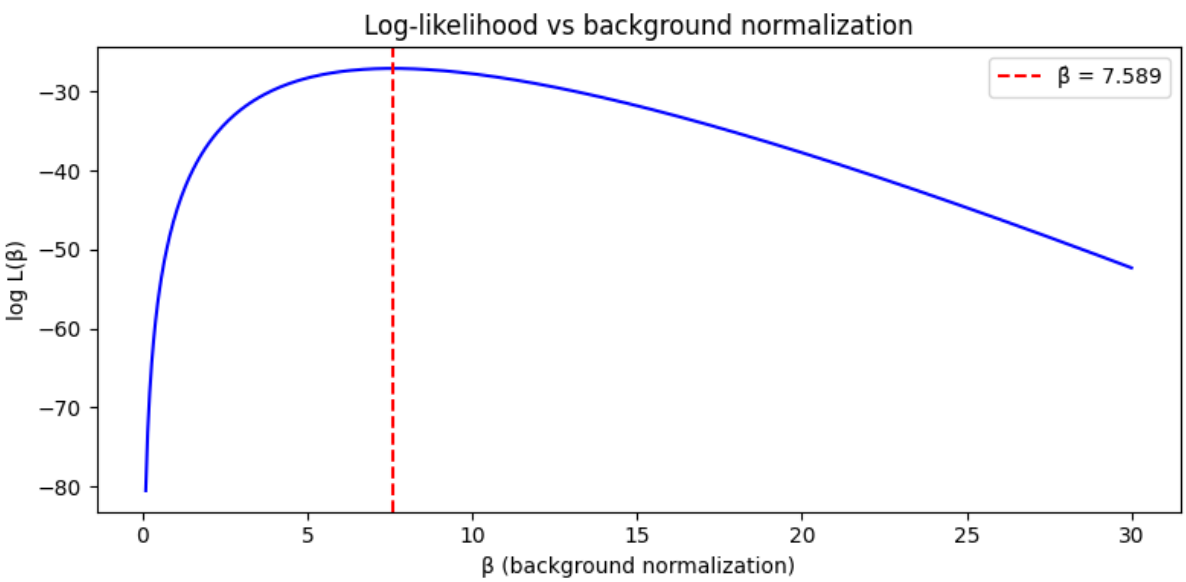

Figure 1: Profile log-likelihood $\ln L(\beta)$ versus background normalization β within the ROI [165.3, 271.7] PE. The X axis denoting the bounds was extended to 30 after preliminary results of setting them up to 5 lead to inconclusive results, as the graph had yet to peak

After the increase of the bounds, the beta was given around the 7.5 range, which is closely matched by later analysis. Such a value indicates that the model provided under-predicts the data, as the nominal background prediction needs to be scaled up by a factor of 7.6.
Then, the Frequentist uncertainty was then calculated with

$$\sigma_\beta \approx \sqrt{\left(-\frac{d^2 \ln \mathcal{L}}{d\beta^2}\Big|_{\hat{\beta}}\right)^{-1}} \tag{22}$$

$\hat{\beta} \pm \sigma_\beta = 7.5887 \pm 1.8972$

7.2 Background fit: the Bayesian perspective

The posterior over the background parameter β was found to peak somewhere around 7-8 for both flat and scale-invariant priors, with both means differing by 0.47, which is small compared to the width of either confidence interval(roughly 12% of the posterior spread).

Prior	Posterior mean $\hat{\beta}$	Confidence interval width
Scale-invariant($\frac{1}{\beta}$)	7.59	3.74
Flat	8.06	3.83

Table 1: Posterior parameter estimates

7.3 Comparing the two approaches

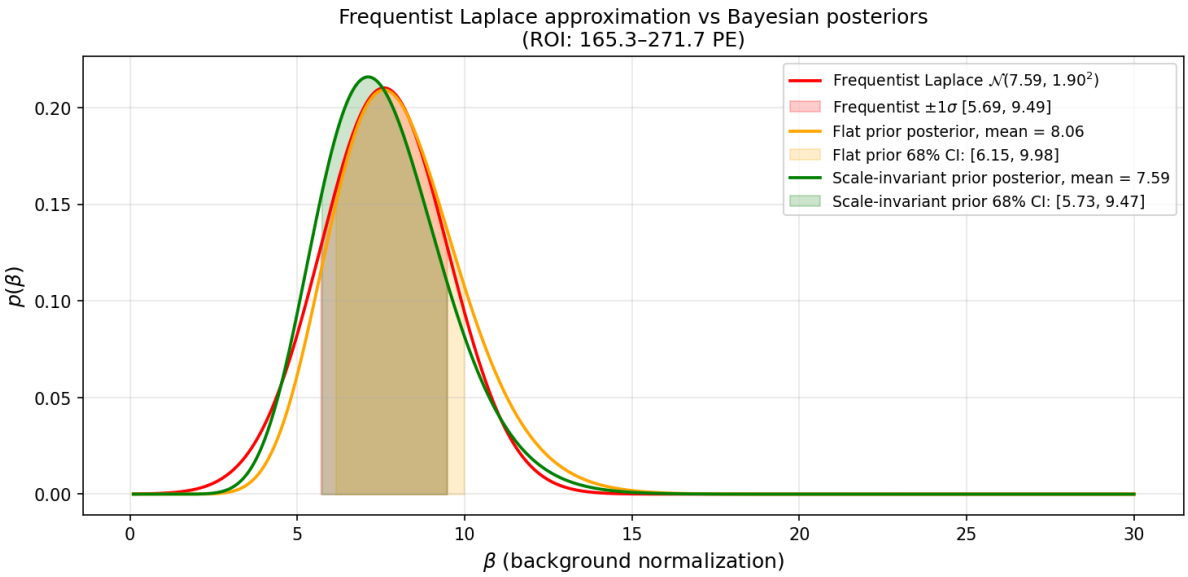

Figure 2: Comparison of Frequentist and Bayesian estimates of the background normalization parameter β within the S2 ROI [165.3–271.7 PE]. This combines the data of Figure 1, Figure 7 and Figure 9. All three estimates are in close agreement, indicating that the posterior is dominated by the data rather than the choice of prior or statistical framework.

7.4 Signal test

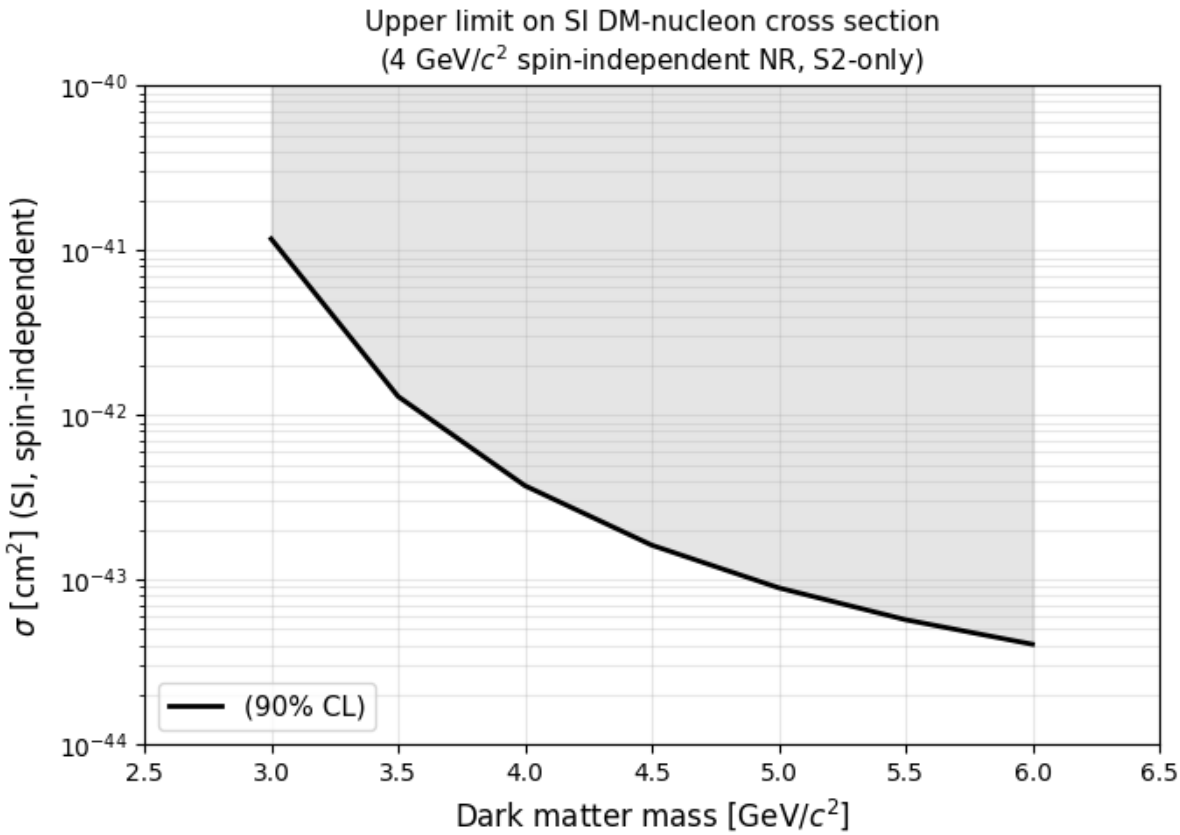

Figure 3: Plot of DM nucleon cross section as a function of dark matter mass. The grey shaded region above the black curve is excluded at 90% confidence rate.

In Figure 3 the curve plotted has a downward slope. This is because the number of expected signal events in the ROI at the (reference) cross section grows with mass. A bigger s_i indicates that even a small μ_s would already predict many events, so μ_s^{up} is reached at a smaller value. Therefore, from Equation 19 we can also conclude a smaller cross section. After implementing the proper code for Equation 21, we find that $\mu_s < 372.1458$ and $\sigma < 3.721 \times 10^{-43}\,\text{cm}^2$ for $m_\chi = 4$ GeV/c^2.

7.5 Bayes factor

To calculate the Bayes factor, first it was necessary to define a joint likelihood function $\mathcal{L}(\text{data} \mid \mu_s, \beta)$ (accounting for both the dark matter signal strength and the background normalization), but now using Bayesian Inference. This way, we could treat both the signal strength μ_s and background normalization β as random variables. However, we used the same Region of Interest used for the Frequentist likelihood (which treated μ_s as a fixed parameter).

Once this likelihood was defined, we calculated the evidences for model 0 (only background) $p(data \mid H_0)$, and model 1 (background+signal) $p(data \mid H_1)$ using equations 9 and 10.
Later, when computing the Bayes factor (through equation 8) we did it for two different priors in order to study the sensitivity to prior: flat (uniform) prior and scale invariant prior (both normalized).

The results of the Bayes factor are shown in the following table:

Prior	Bayes factor B_{01}
Scale-invariant($\frac{1}{\beta}$)	0.0225
Flat	0.8881

Table 2: Bayes factor for different priors

8 Discussion

8.1 ROI and reasoning for cuts

Before being able to properly infer any of the data, it is good to understand like the original report why specific ROIs were important.

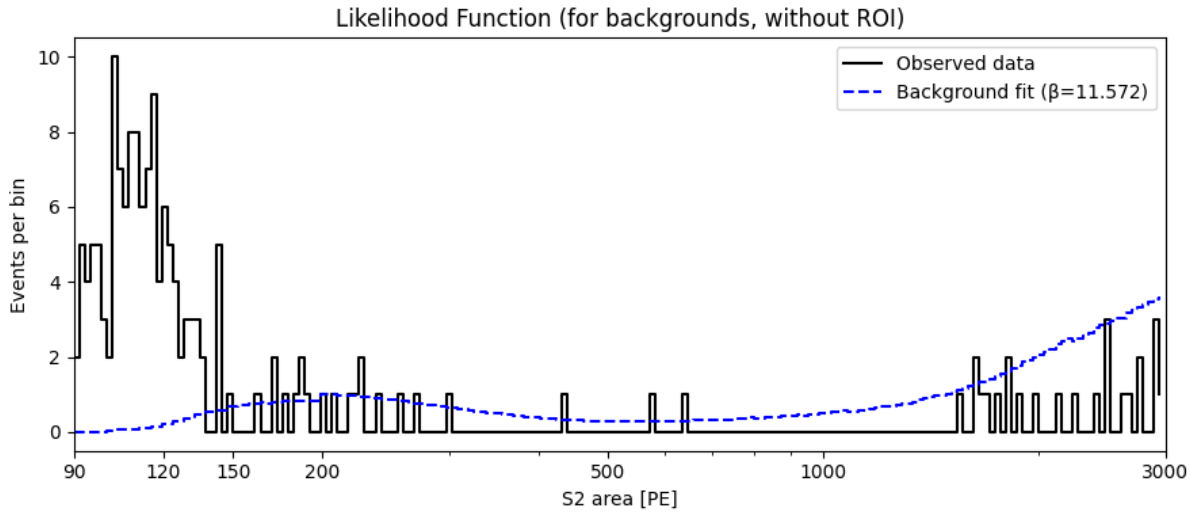

Figure 4: The full data set of the likelihood function. The y axis is the amount of events by the bins defined, and the x axis is the entire breadth of the S2 area, utilizing a logarithmic scale. The dotted blue line indicates the β, which is clearly distorted by the original.

For example, the above figure shows the S2 area from 90 to 3000, and this was after the multiple energy cuts. This is one of the clearest reasons for implementing the ROI that was used in the original report of (165.3 to 271). And, this is evident by figure 2. The background fit of 11.572 is clearly skewed by the values present in the 1000+ range.

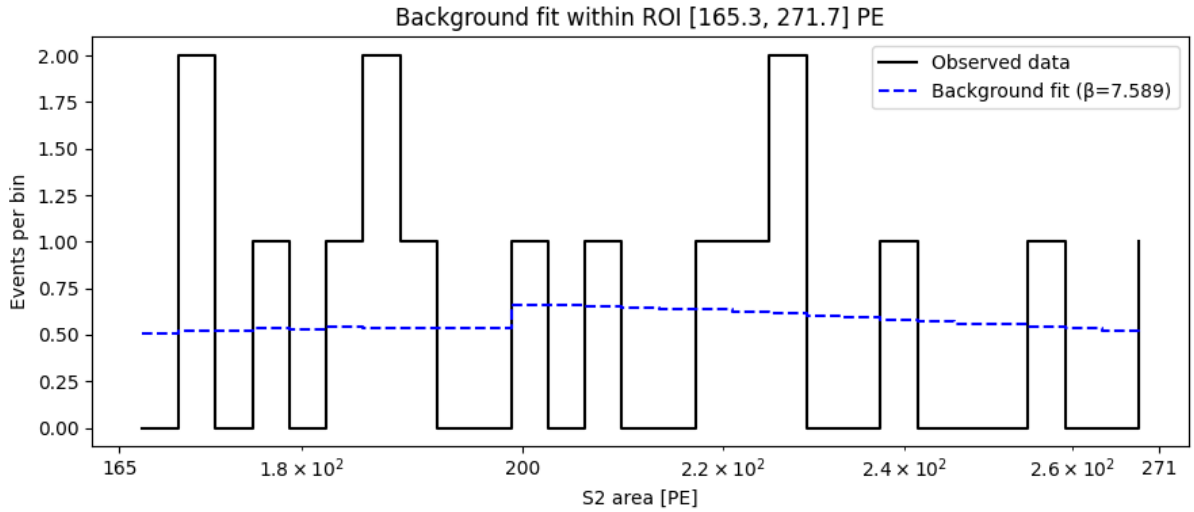

Figure 5: A much smaller extract of Figure 2, focusing on just the ROI of (165.3 to 271). Illustrative to just show the necessity of doing the Log-Likelihood function.

However, when zooming into this ROI, it's clear that working with just likelihood functions yields a quite nonsensical graph, the events per bin and the background fit do not seem to correlate. This is mainly due to there being only 16 events in the ROI, which breaks down the analysis of any smaller functions.

8.2 Background fit

Regarding the results obtained through Bayesian methods, these suggest that inference of β is relatively insensitive to the choice of prior assumptions. Besides, the fact that $\hat{\beta}$ lies far from 1 means the model needs the background to be 7-8 times larger than the nominal prediction to match the counts. However, this is expected since the authors explicitly stated in the data release that the background model is incomplete and conservative([3]).

8.3 Signal test

The value obtained for the upper limit via the Brent algorithm is $\mu_s < 372.1458$. The XENON1T paper doesn't compute a μ_s^{up} value we can compare this to. However, the obtained upper limit of $\sigma^{\text{up}} = 3.72 \times 10^{-43}$ cm^2 differs from the XENON1T paper's value of $\sigma^{\text{up}}_{\text{paper}} \sim 6 \times 10^{-42}$ cm^2 at $m = 4$ GeV/c^2 by a factor of ~ 16. This difference could be attributed to the background models, as in some Figures it is visible they modelled the cathode background, but only made public the ER and CEvNS component csv files 3. Another reason could be that our background model predicts 2.11 events in the ROI, against the 16 observed. Therefore, there is 14 events "unaccounted" for, and the Poisson method doesn't distinguish whether they account for unmodelled background or dark matter signal.

8.4 Bayes factor and prior sensitivity

As we can see in table 2, we obtained quite different results for each prior, but they both have one thing in common: they are lower than 1. This implies that the data observed is more likely under the signal hypothesis (H_1) than the background-only hypothesis (H_0).

Regarding to the prior sensitivity, we can conclude that the evidences of the models (and therefore the Bayes factor) are quite sensitive to the type of prior used. This is because the flat prior penalizes the signal model significantly by spreading probability mass over a large range of signal strengths. It considers that the signal being 1 is as likely as it being 99; however, the excess of events is actually in smaller values, so the flat prior is not fair with the signal values (gives more importance to higher values than it should). On the other hand, the scale invariant prior is more appropriate when the order of magnitude of the signal is unknown. It gives equal weight to every decade (e.g., 0.1 to 1 and 1 to 10), making it more

"neutral". Consequently, it is more fair with small values of the signal, and it notice that the model H_1 fits pretty good the events observed. That is why the scale invariant prior favors much more the model with the dark matter signal over the background-only model.

However, the dramatically low value for B_{01} in the Scale invariant prior case could indicate us that our background model is too rigid. The lack of flexibility could prevent the background from adapting to spectral fluctuations, forcing the Bayesian evidence to favor the model with the dark matter signal just because the background-only model can not properly adapt to the data.

9 Conclusion

This project has implemented a binned Poisson likelihood in order to re-analyze public data from the XENON1T S2-Only research for light dark matter. The analysis has been conducted around the need for the S2-only approach, which rendered the scintillation signal S1 unusable. This has achieved a lower energy threshold that permits the detection of relatively low mass particles suspected to be DM. By focusing on the specific ROI and applying event selections, we aimed to distinguish possible dark matter signals from similar background noise defined by electronic recoils, solar neutrinos and cathode decays.
The comparison between Bayesian and Frequentist frameworks highlights a strong consistency. The Frequentist likelihood peaks at $\beta = 7.59$, a value resembling the Bayesian posterior mean under both flat $\beta = 8.06$ and scale-invariant $\beta = 7.59$ priors. This convergence indicates that within the chosen ROI, the dataset remains partly invariant towards the different statistical model.
The limitations explaining the existing discrepancies between the two β values are due to the nature of the mathematical model. The Frequentist approach only considers the peak for a single value of β. However the Bayesian flat prior method considers the tail distribution of the data which is more significant on the high end. Moreover, from the signal test we can conclude that the data is inconclusive for a DM signal, as the excess of events in the ROI could be attributed to unmodeled background components. We find that for a mass $m_\chi = 4$ GeV$/c^2$, the strength factor is $\mu_s < 372.15$ and the cross section $\sigma < 3.72 \times 10^{-43}$ cm^2.

Author Contributions

- **Nora Andrade Hernandez:** Wrote the Introduction section and the signal analysis for the upper limit (μ_s^{up}), computed μ_s^{up} and the corresponding theory, code (github repository), results, discussion and conclusion sections.

- **Maley Diaw Corral:** Wrote the Data, Model and Statistical Methods(with James) sections as well as half of the Inference Procedure(Background normalization). Regarding the code, worked mainly on the background fit as well as on setting up the likelihood(first parts of the likelihood.ipynb document). Also drafted the Abstract.

- **Tudor Nicolae Vintila:** Repo hygiene and frequentist analysis: As seen in the github commits (QV), the latter part of the likelihood implementation (including log likelihood) and frequentist analysis, plus explanations of why RoI was chosen, which were later added to the report (Mainly in section 8.1). After a swap from working from 1 jupyter file to multiple, the repo was reworked to now have a loader function and multiple .ipynb files, creating the background comparison and made the readme explaining that.

- **James Conlon:** In terms of writing the report: physical Background, briefly results, discussion, conclusion with Nora, Theory for Bayesian inference and the introductory and part of the statistical methods with Maley, code for posterior distribution with credible interval.

- **Ana Alonso Hernando**: In terms of code, worked on defining a new likelihood for Bayesian inference and calculating evidences for the two models. Also, calculated Bayes factor for two different priors. Regarding to the writing in the report, added the correspondent parts of Bayes factor in the following sections: statistical methods, results and discussion.

A Additional Figures

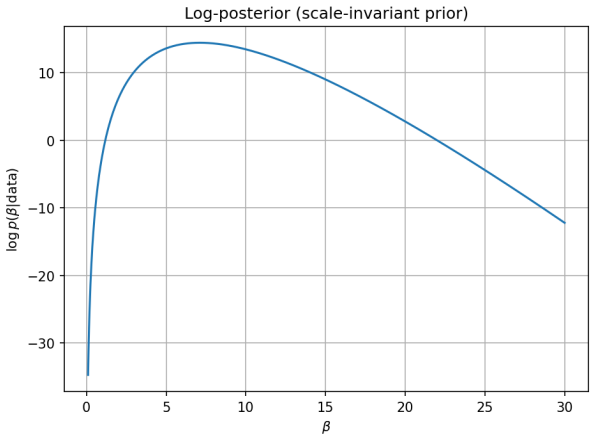

Figure 6: Log-posterior of the background normalization β under a scale-invariant prior.

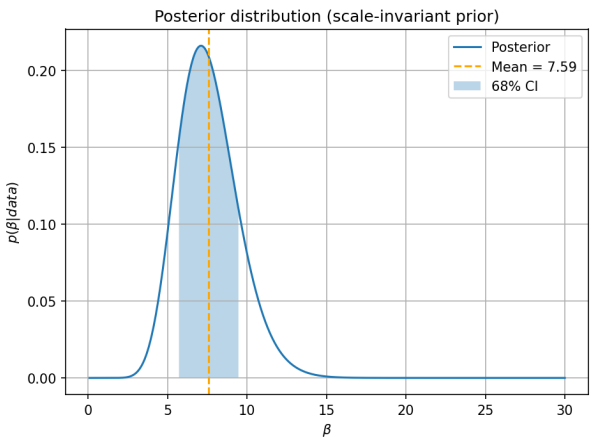

Figure 7: Posterior distribution of β under a scale-invariant prior, with mean and 68% credible interval.

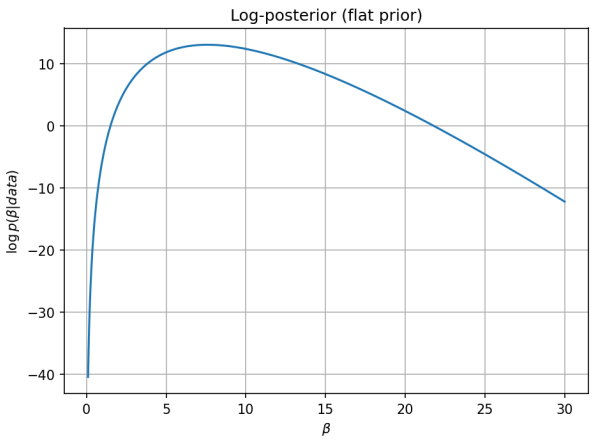

Figure 8: Log-posterior of the background normalization β under a flat prior.

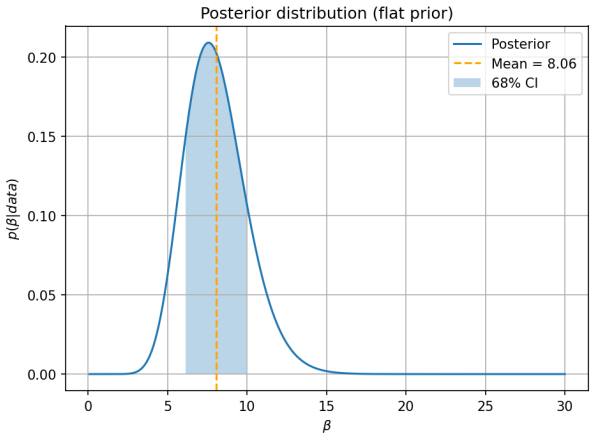

Figure 9: Posterior distribution of β under a flat prior, with mean and 68% credible interval.

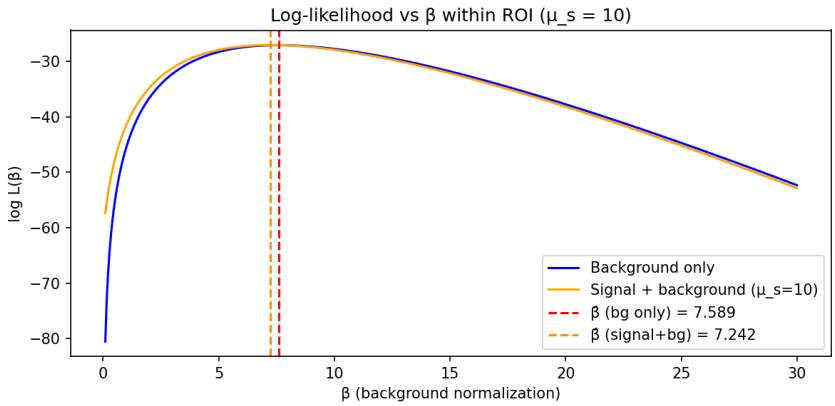

Figure 10: Log-likelihood as a function of β for the background-only and signal-plus-background models within the ROI. Note, the mu_s chosen here is to showcase the comparison, but the value 10 here is arbitrary

B Reproducibility

- software used
  Visual Studio Code, across Mac and Windows.

- package versions
  numpy==2.4.3
  pandas==3.0.1
  scipy==1.17.1
  matplotlib==3.10.8
  wimprates==0.5.0
  jupyter==1.1.1

- instructions for running the code
  See README of report

- link to GitHub repository
  `https://github.com/QuadroVert/DM1StatsGroupRepo2026.git`

C References

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